CLOUD & CYBERSECURITY · RESEARCH + PROTOTYPE

Risk-Adaptive Zero-Trust Security Architecture for Hybrid Cloud Infrastructure

Farhaan Pasha · SRN R23EK009

Profile

STUDENT

Farhaan Pasha

SRN

R23EK009

FACULTY GUIDE

Mr Hidam Rameshwar Singh

CENTER

Center for Emerging Computing Technologies

Project Description

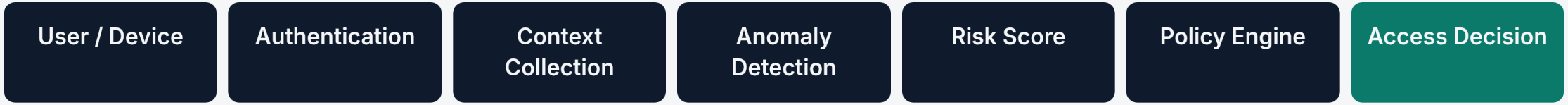
Hybrid clouds span private and public resources, so users, devices and apps cross multiple trust boundaries.

Core idea: evaluate every access request continuously, assign a risk level, and apply an adaptive decision.

Objectives

- Design a Zero-Trust hybrid cloud architecture
- Detect behavioural anomalies
- Build risk-adaptive access control
- Evaluate accuracy, false positives, overhead, latency
- Test against selected attack scenarios

System Workflow and Decisions



Low	Medium	High	Critical
Allow	Challenge	Restrict	Block

Signals: identity and privileges, device, behaviour score, resource, request, security context.

Literature Review

Zero Trust foundations

NIST SP 800-207 (Rose et al., 2020) defines Zero Trust Architecture; Google's BeyondCorp (Ward and Beyer, 2014) drops trust based on network location.

Anomaly detection

Chandola et al. (2009) survey the field; Isolation Forest (Liu, Ting and Zhou, 2008) is a common unsupervised baseline for behavioural data.

Risk-adaptive access control

The JASON report (2004) proposed risk-based access decisions in place of fixed static rules.

Gap: few prototypes combine all three on a hybrid cloud with measured latency and false-positive cost.

Expected Outcomes

- Hybrid cloud prototype with Zero-Trust access control
- Behavioural anomaly scoring with Allow / Challenge / Restrict / Block decisions
- Quantitative results: detection accuracy, false-positive rate, authentication overhead, service latency
- Technical report, source code and final presentation

Build · Detect · Assess · Adapt · Secure